

# ENVIZ: DreamWorks GPU-Accelerated Interactive Environment Deformation and Visualization Toolset

Arunachalam Somasundaram  
DreamWorks Animation  
California, USA  
arun.somasundaram@dreamworks.com

Chris Michael  
DreamWorks Animation  
California, USA  
chris.michael@dreamworks.com

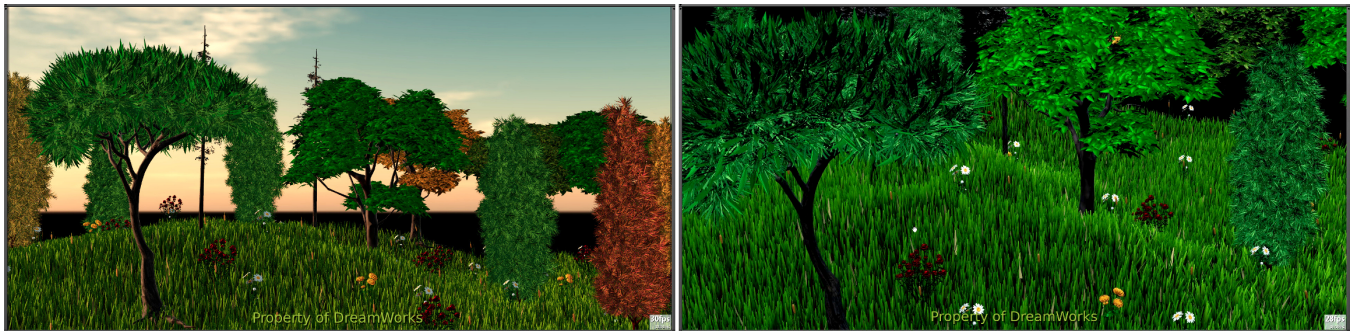


Figure 1: Environment deformed and visualized using ENVIZ interactively in the 3-D viewport.

## ABSTRACT

ENVIZ is a toolset at DreamWorks Animation that uses the power of the GPU for both interactive deformation and visualization of environments in the 3-D viewport. It replicates the final rendered hi-fidelity motion, and reacts to material shaders and lights to produce a well shaded and lit scene. Meshes with millions of points can be deformed and visualized real-time in the viewport which can help the artist evaluate motion for a shot without having to run costly renders thus providing significant time and resource savings.

### ACM Reference Format:

Arunachalam Somasundaram and Chris Michael. 2024. ENVIZ: DreamWorks GPU-Accelerated Interactive Environment Deformation and Visualization Toolset. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH Talks '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641233.3664324>

## 1 MOTIVATION

Environments in DreamWorks films encompass a variety of vegetation assets such as grass, plants, trees etc. A shot in films such as *Puss in Boots: The Last Wish* or *The Wild Robot* could have tens or hundreds of environment assets, which could translate to millions of polygons [Figure 1] that need to be deformed and visualized to evaluate motion. Given the large amount of data involved and the iterative nature of shot work, evaluating environment motion in the 3-D viewport using CPU-based methods is prohibitively slow.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Talks '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0515-1/24/07

<https://doi.org/10.1145/3641233.3664324>

Also, exporting large amounts of data to disk and rendering the shot to evaluate motion is laborious and requires render resources.

ENVIZ replicates the same hi-fidelity deformation that is performed on the CPU, instead utilizing the GPU to accurately deform and visualize results interactively in the 3-D viewport. It reacts to materials and lighting allowing the artist to evaluate motion using lighting conditions similar to those used in an actual render. While the use of vertex shaders for real-time procedural foliage animation is common within the games industry, ENVIZ incorporates GPU accelerated displays into our existing workflow architecture without the need to render or output to a game-engine which may approximate or produce a different result than in our renders.

While DreamWorks uses GPU for approximating hair curves' deformation and display using WIZ [Somasundaram and Shay 2022], ENVIZ's broader scope allows for the true deformation and display of polygonal and instanced geometry for entire environments.

## 2 ENVIRONMENT MOTION

Artist-driven environment motion at DreamWorks can be largely classified into two categories, namely, i) Curve-driven Motion, and ii) Transform-driven Motion. These categories are described next.

### 2.1 Curve-driven Motion

For polygon geometries such as plants, tree branches, or grass, a curve rig [Figure 2] is first created [Michael and Somasundaram 2020] by the artist which is then used to drive the polygonal geometry. The curve rig is derived from the geometry, and can be hierarchical in nature. The curves typically pass through the central core of the polygonal geometric structure. We use a NURBS type (cubic order) for the curves. Every point on the polygon is driven by a curve using a custom polygon-to-curve wrap that uses smooth binding, i.e. a polygonal point is bound and driven by 4 curve vertices weighted using the NURBS basis weights.

## 2.2 Transform-driven Motion

Several of the environment assets such as leaves are instances of prototypes. They are created by instancing prototype geometries (such as different types of leaves) at the corresponding specified point locations with varying scales and rotations. There could be hundreds of thousands of those instances per environment asset (for e.g., leaves per tree). The motion of these instanced geometries is controlled by manipulating their position, scale, and rotation.

## 3 ENVIZ CACHE

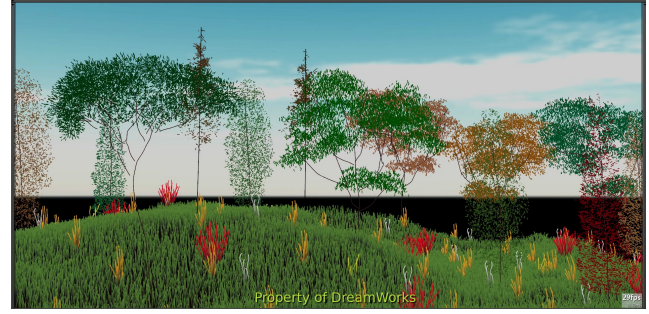
ENVIZ first creates a single ENVIZ Cache at reference pose for each environment asset that needs to be deformed. This cache is stored on disk as an Alembic file and is used later during deformation and display of environment motion using the GPU. For environment assets that need curve-driven motion, the ENVIZ Cache contains the polygonal geometry, the curve rig, and binding data between both of them. The binding data contains binding information for each polygonal point  $P_i$ , namely, i) the bound-to curve  $C_{P_i}$ , ii) the four vertices  $V_{P_i}^j$  ( $0 \leq j \leq 3$ ) of  $C_{P_i}$  that  $P_i$  is smooth-bound to, iii) the four binding weights  $W_{P_i}^j$  ( $0 \leq j \leq 3$ ) (those are the NURBS basis weights of  $P_{closest}$ , which is the point on  $C_{P_i}$  that is closest to  $P_i$ ), iv) the parametric  $U$  of  $P_{closest}$  on  $C_{P_i}$ , and v) the offset  $O_{P_i}$  of  $P_i$  from  $P_{closest}$  along the local axes of  $C_{P_i}$  at  $P_{closest}$ . This binding calculation is performed once per asset at reference pose on the CPU. For transform-driven geometry, the cache stores the prototype geometries, and point instancing data such as position, scale, rotation matrix, and the corresponding prototype numbers.

## 4 ENVIZ GPU DEFORMATION

ENVIZ uses a custom GLSL Vertex Shader to perform the deformation at interactive speeds on the GPU. ENVIZ is designed to minimize the data transfer from the CPU to GPU for increased speed. Data from ENVIZ Cache is read in once. For curve-driven assets, the input per frame to ENVIZ deformation is just the animated curves's vertex positions and local axes. For each such environment asset, the binding information obtained from the ENVIZ cache is used to deform that asset using a custom polygon-to-curve wrap implemented in the GLSL Vertex Shader. The deformed position of point  $P_i$  is  $Pos_i$ , where  $Pos_i = \sum_{j=0}^3 W_{P_i}^j * (Pos_{V_{P_i}^j} + O_{P_i} * Axis_{V_{P_i}^j})$ .  $Pos_{V_{P_i}^j}$  and  $Axis_{V_{P_i}^j}$  are the corresponding position and local axes of  $C_{P_i}$  at  $V_{P_i}^j$ . For transform driven instances, the per frame deformation input is a set of positions, scales, and rotation matrices. ENVIZ instances the geometric prototypes in GLSL for increased speed, i.e., for each prototype that is instanced, only one call is made from the CPU to the GPU passing all the corresponding instanced points' transform data. Once the vertices are deformed using the GLSL Vertex Shader, the GLSL Geometric Shader constructs the polygonal geometry from the vertices, calculates new normals, and passes it to the GLSL Fragment Shader.

## 5 ENVIZ GPU VISUALIZATION

ENVIZ uses a custom GLSL Fragment Shader that takes the incoming geometry and shades it according to the assigned materials and lighting. It also responds to environment lights making it easier to



**Figure 2: Environment Deformation Rig: Curves (Plants / Grass / Tree branches) and Transforms (Leaves)**

light the scene. Artists can interactively change lighting and material properties and see an immediate response in the 3-D viewport.

## 6 IMPLEMENTATION AND INTEGRATION

ENVIZ is implemented using custom nodes in a third-party package. The *ENVIZ Cache* node creates the Alembic cache at reference pose for both curve-driven and transform-driven geometries. The *ENVIZ Display* node deforms and displays the geometry using the GPU. ENVIZ is integrated into Grasshopper [De St. Jeor et al. 2019] which is DreamWorks' environmental motion system. With this integration, artists have access to improved playback speed, allowing motion and performances to be quickly assessed.

## 7 RESULTS AND CONCLUSION

ENVIZ is a highly efficient toolset that uses the power of the GPU to interactively deform and visualize well shaded and lit environments in the 3-D viewport which saves a lot of time and resources for the artist during development and shot work. As an example, the playback of a shot containing 21 million points (corresponding to 15 million primitives) across multiple assets had a conventional playback of  $\approx 1$  frame per second using CPU (64 cores - 128 threads running in parallel). With ENVIZ, this playback was improved to  $\approx 27$  frames per second using GPU (10,752 CUDA cores). This difference becomes more pronounced as point count increases, with ENVIZ performing real-time while the CPU-based mode becomes slower. Because this playback is both fast and visually representative of the final output, shots require fewer render iterations - saving render resources and cutting down significant wait times between versions. ENVIZ is being used by Character Effects artists working on environment motion at DreamWorks Animation.

## REFERENCES

- Chris De St. Jeor, Chris Michael, Kurt Phillips, and Arunachalam Somasundaram. 2019. Grasshopper: Dreamworks Environmental Motion System. In *ACM SIGGRAPH 2019 Talks* (Los Angeles, CA) (SIGGRAPH '19). Association for Computing Machinery, New York, NY, USA, Article 62, 2 pages. <https://doi.org/10.1145/3306307.3328174>
- Chris Michael and Arunachalam Somasundaram. 2020. Termite: DreamWorks Procedural Environment Rigging Tool. In *ACM SIGGRAPH 2020 Talks* (Virtual Event, USA) (SIGGRAPH '20). Association for Computing Machinery, New York, NY, USA, Article 9, 2 pages. <https://doi.org/10.1145/3388767.3407384>
- Arunachalam Somasundaram and Kelly Shay. 2022. WIZ: DreamWorks GPU-Accelerated Interactive Hair/Fur Deformation and Visualization Toolset. In *ACM SIGGRAPH 2022 Talks* (Vancouver, BC, Canada) (SIGGRAPH '22). Association for Computing Machinery, New York, NY, USA, Article 54, 2 pages. <https://doi.org/10.1145/3532836.3536245>